# Special Issues 1 & 2 (PDF)

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# Special Issue 1

## What happens when an EC2 instance stops running?

When an Amazon EC2 instance is stopped or terminated, several things can happen depending on the configuration of the instance and the type of storage attached to it.

First, if an EC2 instance is **stopped**, the virtual machine itself shuts down, but the **attached EBS storage volume is preserved**. This means that files stored on the EBS volume (such as application code, Docker containers, and configuration files) remain intact. When the instance is started again, the system boots from the same EBS volume, and the data is still available.

However, if an EC2 instance is **terminated**, the instance is permanently deleted. If the attached storage was configured to delete on termination (which is the default for many instances), the data will also be lost. Because of this, it is important to create backups of important instances.

To prevent data loss and preserve a working server environment, AWS allows users to **create an Amazon Machine Image (AMI)**. An AMI is essentially a snapshot of the EC2 instance that includes:

- The operating system
- Installed software
- Configuration files
- Attached storage volumes

This AMI can then be stored in AWS and used to recreate the instance at any time.

A common workflow is:

1. Create an EC2 instance and configure it.
2. Install software such as Docker, Nginx, or application dependencies.
3. Once the instance is fully configured, create an **AMI snapshot**.
4. Store the snapshot in AWS (internally backed by Amazon S3).
5. If the instance fails or is terminated, launch a **new EC2 instance from the AMI**.

This effectively allows the instance to be **"frozen" and duplicated**, making it possible to restore the server or deploy multiple identical environments.

### Costs involved

There are several costs associated with this process:

**1. EBS Snapshot Storage**
When an AMI is created, AWS stores snapshots of the attached EBS volumes. These snapshots are stored in Amazon S3 internally and are billed based on storage size.

Typical pricing is approximately:

- Around **$0.05 per GB per month** for snapshot storage.

For example, if the instance has an 8 GB disk, the monthly snapshot storage cost would be very small (roughly a few cents per month).

**2. EC2 Instance Costs**
If the instance is running, compute charges apply depending on the instance type. In the learner lab environment, the instance type used is typically **t2.micro**, which is included in the free tier or covered by lab credits.

**3. AMI Storage Costs**
AMIs themselves do not incur high costs, but the underlying snapshots stored in S3 do.

Overall, the cost of storing an AMI for a small instance is minimal and is a common practice for backup and disaster recovery.

# Special Issue 2

## What happens when an EC2 instance is rebooted, and what are the options for publishing a public URL?

When an EC2 instance is rebooted or stopped and started again, the **public IP address assigned to the instance may change**. This happens because the default public IP address in AWS is **dynamic**, meaning AWS assigns a temporary IP address each time the instance starts.

This can cause a problem if the website is publicly accessible and the URL has already been shared with users. For example, if a website was originally hosted at:

```
http://54.xxx.xxx.xxx
```

And the instance restarts, AWS might assign a completely different IP address. As a result, the previously shared link will no longer work.

### Options to keep a stable public address

There are several solutions to this problem.

#### 1. Elastic IP Address

The most common solution is to use an **Elastic IP (EIP)**.

An Elastic IP is a **static public IPv4 address** that is allocated to an AWS account. Unlike normal EC2 public IP addresses, an Elastic IP does not change when the instance is stopped or restarted.

The workflow is:

1. Allocate an Elastic IP in AWS.
2. Associate the Elastic IP with the EC2 instance.
3. Access the server using the Elastic IP address.

This ensures the server always remains accessible through the same IP address.

#### 2. Domain Name + DNS

Another option is to use a **domain name** with a DNS service such as Amazon Route 53.

Instead of sharing the raw IP address, users access the website through a domain such as:

```
https://visionforge.ai
```

The DNS record then points to the current server IP. If the server changes, the DNS record can be updated without changing the public URL.

### Costs involved

Elastic IP addresses are mostly free **while attached to a running instance**.

However, AWS charges small fees in certain situations:

- If an Elastic IP is **allocated but not attached** to an instance.
- If more than **one Elastic IP** is used per instance.

Typical pricing is approximately:

- Around **$0.005 per hour** for unused Elastic IP addresses.

DNS services such as Route 53 also have small monthly fees depending on hosted zones and DNS queries.

### Can this be done in the Learner Lab environment?

In many AWS Learner Lab environments, users have limited permissions. In some cases:

- Elastic IP allocation may be **restricted or disabled**.
- Certain networking features may not be available.

However, in many lab environments, it is still possible to allocate and associate a single Elastic IP address with an EC2 instance. If it is not allowed, the only available option is to **use the temporary public IP address assigned by AWS**.

In a production environment outside of a lab, using an Elastic IP or a domain name with DNS is the recommended approach for maintaining a stable public URL.